

UTSIRA FYR, Norway

WMO: 014030

Lat: 59.306N

Lon: 4.872E

Elev: 185

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.3	27.5	10.3	9.4	27.4	15.1	11.9	29.9	50.9	37.7	46.9	37.1	11.7	110	0.887

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	69.7	62.6	66.9	61.3	64.6	59.9	63.8	68.1	62.2	65.8	60.7	63.8	12.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	83.6	65.8	60.5	79.2	64.5	59.1	75.2	62.9	29.2	68.3	28.0	66.0	27.0	63.9	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
45.4	40.6	36.2	DB	23.6	74.5	4.2	4.3	20.6	77.6	18.1	80.1	15.8	82.5	12.8	85.7
			WB	20.7	65.3	3.8	2.8	18.0	67.4	15.7	69.1	13.6	70.7	10.8	72.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	38.0	36.6	38.1	42.7	48.4	53.3	58.2	59.5	55.6	49.1	43.5	40.0	
	DBStd	8.98	4.49	4.71	3.98	3.81	4.42	3.64	4.15	3.82	3.64	4.15	4.70	5.27	
	HDD50	2002	371	375	369	222	83	7	0	0	3	63	198	311	
	HDD65	6589	836	795	834	668	515	350	218	176	284	492	645	776	
	CDD50	903	0	0	0	4	34	107	254	296	170	36	2	0	
	CDD65	15	0	0	0	0	0	1	6	7	1	0	0	0	
	CDH74	15	0	0	0	0	1	0	9	5	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	18.0	22.7	19.8	17.8	15.9	16.1	15.1	14.5	15.9	18.0	20.1	20.2	20.2	
Precipitation	PrecAvg	54.7	5.5	4.3	4.0	3.1	2.4	2.8	3.5	5.0	5.4	6.2	6.4	5.9	
	PrecMax	64.8	11.6	8.3	7.2	5.2	4.2	6.0	6.6	7.8	8.4	11.1	12.0	11.1	
	PrecMin	38.1	1.1	1.4	0.3	1.3	0.3	0.8	1.6	1.3	1.9	2.1	1.9	1.9	
	PrecStd	7.3	2.6	1.8	1.8	0.9	0.9	1.2	1.3	1.4	1.8	2.4	2.4	2.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.3	45.7	49.2	59.2	67.5	69.4	74.8	73.3	67.3	58.9	53.8	49.9	
		MCWB	45.6	44.0	42.9	50.3	56.0	59.5	64.0	64.4	60.1	55.3	52.0	48.2	
	2%	DB	45.8	44.5	44.9	53.2	61.2	64.0	70.0	69.4	64.0	56.9	51.9	48.6	
		MCWB	44.1	43.2	42.4	46.6	53.2	57.1	62.9	63.5	58.5	53.8	49.8	47.0	
	5%	DB	44.9	43.7	43.8	49.7	57.7	60.6	66.6	67.0	61.9	55.5	50.6	47.6	
		MCWB	43.3	42.5	41.8	45.3	51.2	55.7	61.6	62.0	57.5	53.1	48.4	45.9	
	10%	DB	43.9	42.8	42.8	47.5	54.6	58.3	64.1	64.9	60.4	54.4	49.3	46.5	
		MCWB	42.2	41.7	41.2	44.2	50.1	54.5	60.1	60.7	56.6	51.9	47.1	44.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	44.9	44.3	51.2	57.8	60.7	65.7	67.1	61.8	56.4	52.5	48.6
			MCDB	46.8	45.2	45.6	56.7	64.7	66.3	71.5	70.2	65.2	57.7	53.2	49.2
2%		WB	44.8	43.8	43.2	47.7	54.1	58.0	63.8	64.3	59.9	54.8	50.4	47.6	
		MCDB	45.6	44.3	44.2	50.8	59.5	62.6	68.5	68.5	62.8	56.0	51.6	48.4	
5%		WB	43.9	42.9	42.5	46.1	52.0	56.4	62.2	62.8	58.5	53.8	49.1	46.4	
		MCDB	44.8	43.5	43.4	49.0	56.4	59.8	66.1	66.2	61.2	55.2	50.2	47.4	
10%		WB	42.8	41.9	41.5	44.7	50.4	55.0	60.5	61.2	57.1	52.6	47.7	45.2	
		MCDB	43.8	42.7	42.6	46.8	53.9	57.8	63.6	64.2	59.9	54.1	49.0	46.4	
Mean Daily Temperature Range	5% DB	MDBR	4.6	4.6	5.1	6.1	7.0	6.6	6.3	6.0	5.3	4.6	4.5	4.6	
		MCDBR	5.1	4.4	5.7	9.3	11.9	9.5	9.7	8.1	6.5	4.9	4.9	4.8	
		MCWBR	5.8	4.7	4.6	5.9	6.6	5.8	5.8	5.1	4.4	4.9	5.7	5.7	
	5% WB	MCDBR	4.9	4.2	4.9	8.1	10.4	8.3	8.8	7.3	5.7	4.8	5.1	4.8	
		MCWBR	5.8	4.6	4.6	5.6	6.4	5.7	5.8	5.2	4.7	5.3	6.2	5.8	
Clear-Sky Solar Irradiance	taub	0.313	0.336	0.354	0.382	0.366	0.362	0.376	0.376	0.361	0.340	0.314	0.290		
	taud	2.162	2.280	2.341	2.310	2.427	2.473	2.451	2.448	2.467	2.398	2.229	2.102		
	Ebn at Noon	146	212	247	260	274	277	269	260	245	210	147	109		
	Edh at Noon	15	22	28	34	33	32	32	30	25	20	14	11		
All-Sky Solar Radiation	RadAvg	87	267	645	1189	1641	1756	1539	1186	756	365	124	55		
	RadStd	13	45	83	108	177	206	146	108	65	45	16	7		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-211	N/A	N/A

Nomenclature: See separate page